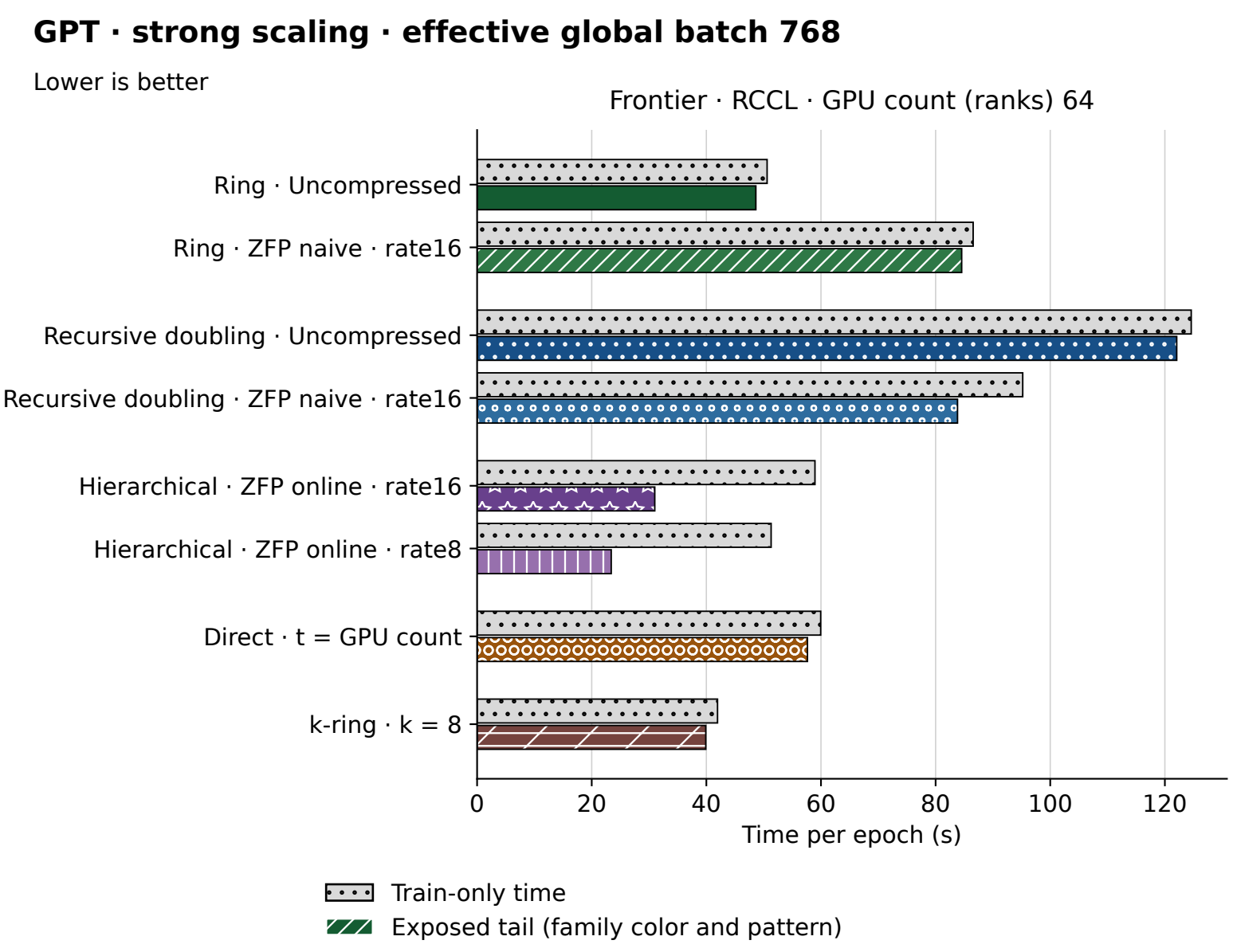

GPT · strong scaling · effective global batch 768
Lower is better
Frontier · RCCL · GPU count (ranks) 64
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
Direct · t = GPU count
k-ring · k = 8
0
20
40
60
80
100
120
Time per epoch (s)
Train-only time
Exposed tail (family color and pattern)
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate